

NHS Services Analysis: infrastructure Utilisation and Healthcare Capacity

The central role of this analysis is to refine healthcare into actionable analytical insights based on the available NHS data. The stakeholder working within the NHS is seeking clarity trend of the healthcare infrastructure and staff resources. These analysis are aimed at guiding decisions between expanding or optimising existing structures. Our analysis focuses on two key business questions:

1. Has there been adequate staff and capacity across NHS networks?
2. What is the actual utilisation of NHS healthcare resources?

Our analytical approach involves trend detection, exploratory visualisation to identify trends and inefficiencies in the NHS services during the available months.

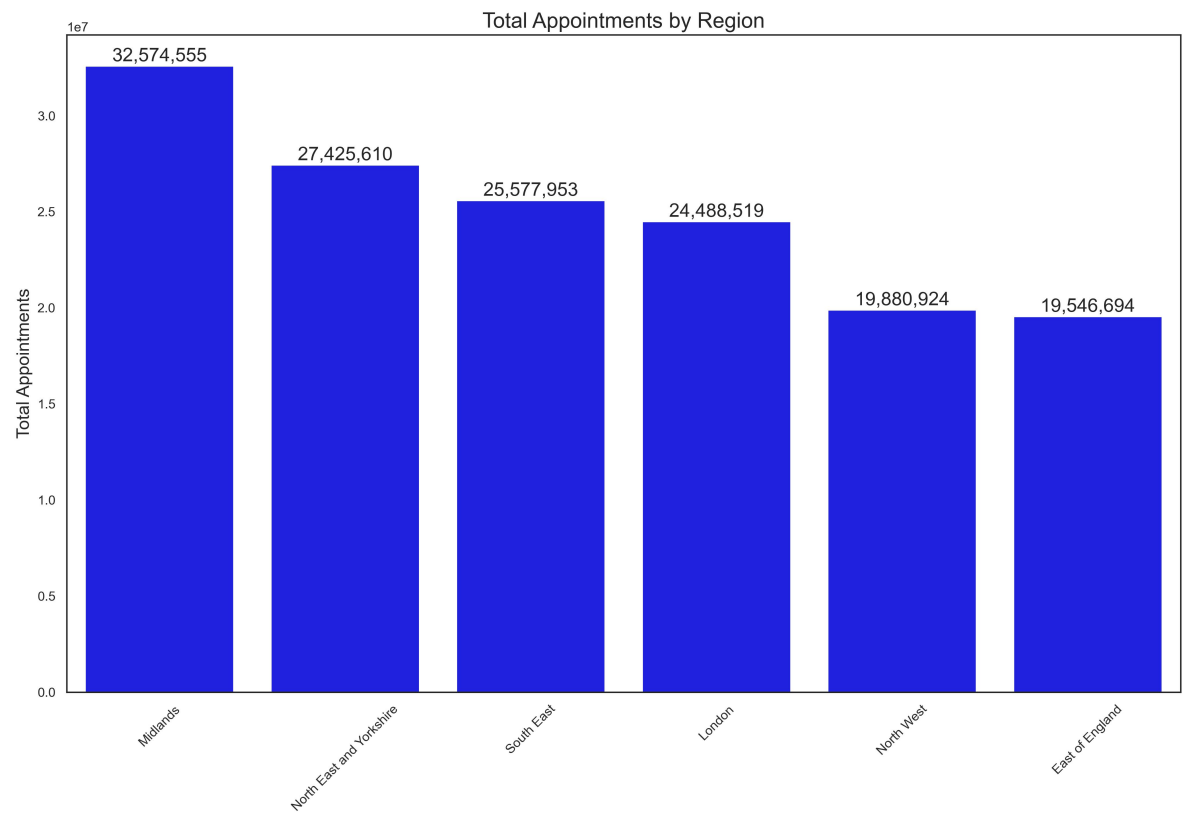
Three main dataset were available (actual_duration, appointment_regional, and national_categories). Each contains a count of appointments column, which make efficient to compare trend between different features. An initial imports of the datasets revealed no missing value but there is a significant amount of unmapped values, which is a limitation to some sector of our analysis as in the “Count of appointment by actual duration” and in the query for a correlation between the time between the booking and the appointment and the appointment status.

In the descriptive analysis we find out that the NHS services data are available across 106 location and 7 regions. I choose the NHS region for a more effectively visualisation (which can be used for further analysis) because the ICB locations are a too numerous too visualise.

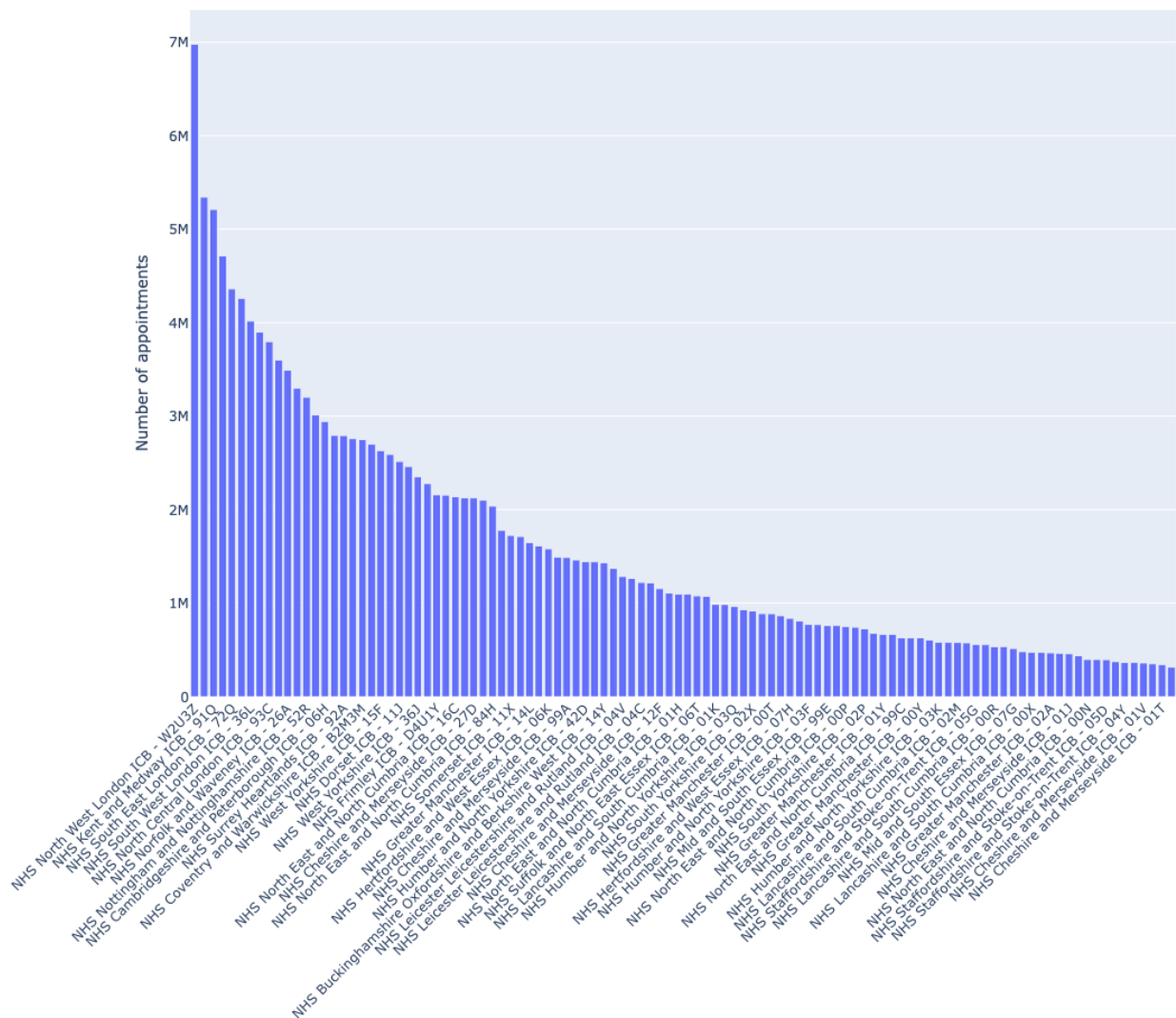
The data were available from August 2021 to June 2022. Data across these months were analyse to identify seasonal trends as staffing challenges, and demand cycles. I tried to detect seasonal trend in the several fields: healthcare professional type, appointment mode, attended/did not attended appointments, context type, NHS service setting and utilisation of NHS facilities.

These choices were maid in the goal of generating meaningful, reproducible insights, while also laying the ground for incorporating regional differentiation in further analysis.

REGION VS ICS LOCATIONS

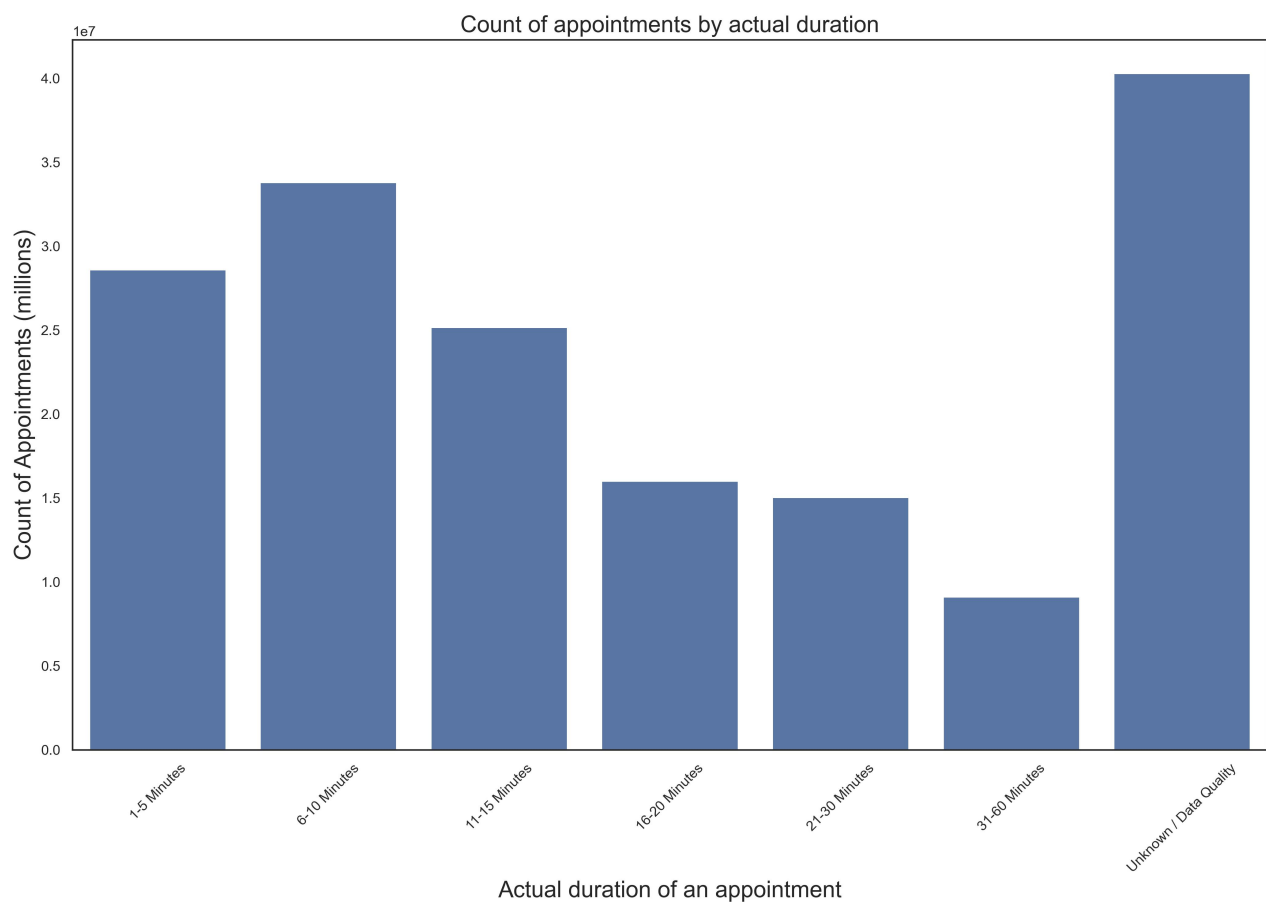


Appointments by ICB Location



The datasets provided were generally complete, with monthly data covering multiple layers of healthcare services and detections. However several limitations were detected:

- unknown or unmapped categories. Many variables (actual_duration, appointment_status and 'time_between_book_and_appointment) contained 'unknown' or 'unmapped entries. These compromised the integrity of certain analyses and suggest an improved data collection protocols

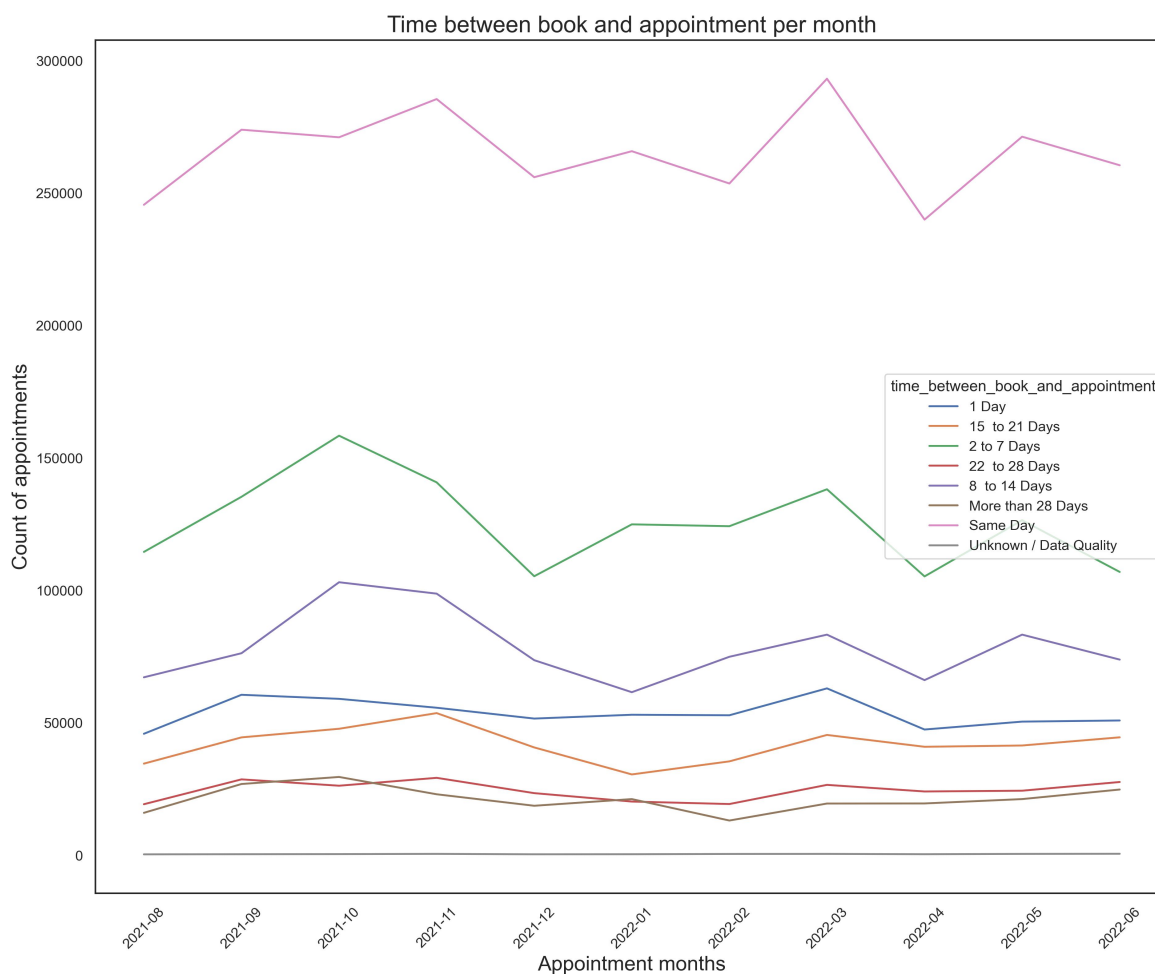


- Temporal gap. While the 11 months available provide insight for seasonal trends, a longer time horizon would be valuable for identifying persistent structural trend

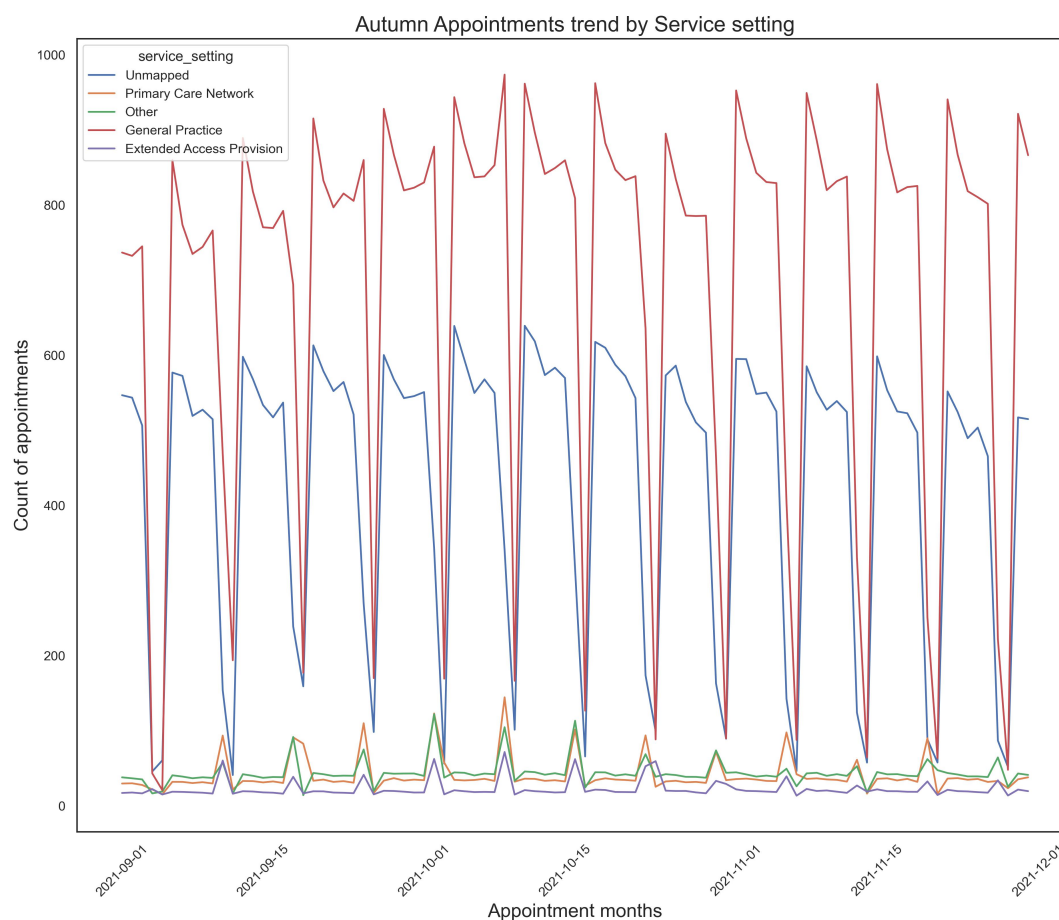
Despite these issues, clear patterns emerged:

- face-to-face appointments are roughly double the volume of telephone appointments, especially during peak period

- Service demand is seasonal. Appointment peak especially in Autumn and in the month of March. Declines are observed during winter months and late spring, suggestion possible correlation to holiday periods and seasonal illness
- The presence of a significant number of DNAs and long wait times between booking and appointments inefficiencies in scheduling especially during the peak period (purple line in the graph - from 8 to 14 days)



The visualisation approach prioritised clarity and comparison through different aspect of NHS services offers. I chose line chart for temporal trends which shows monthly variations in appointment volumes and professional type. Line charts helps visualising seasonal changes in the most common service setting (National category, Appointment mode, Healthcare professional type) and shows that appointment volume swing significantly, which can be especially detect from the seasonal charts.



A histogram of actual duration shows that most of the appointments lasted between 1 to 15 minutes. However a significant number of an appointments were classified as ‘unknown/data quality’.

I decided to use a utilisation coefficient to compare the usage of available appointments against a benchmark of 1.2 million daily appointments. By displaying both the number of daily appointments and the utilisation coefficient on the same chart. The fact that the red and blue lines follow the same trajectory confirms that the benchmark is valid and we can observe whether NHS facilities are being over-utilised or underutilised.

Each of the visualisation was refined by adjusting axis labels, adding legends, and formatting title to ensure the message could be quickly and easily understood at a glance.

JUPYTER NOTEBOOK AND USE OF TWITTER DATA

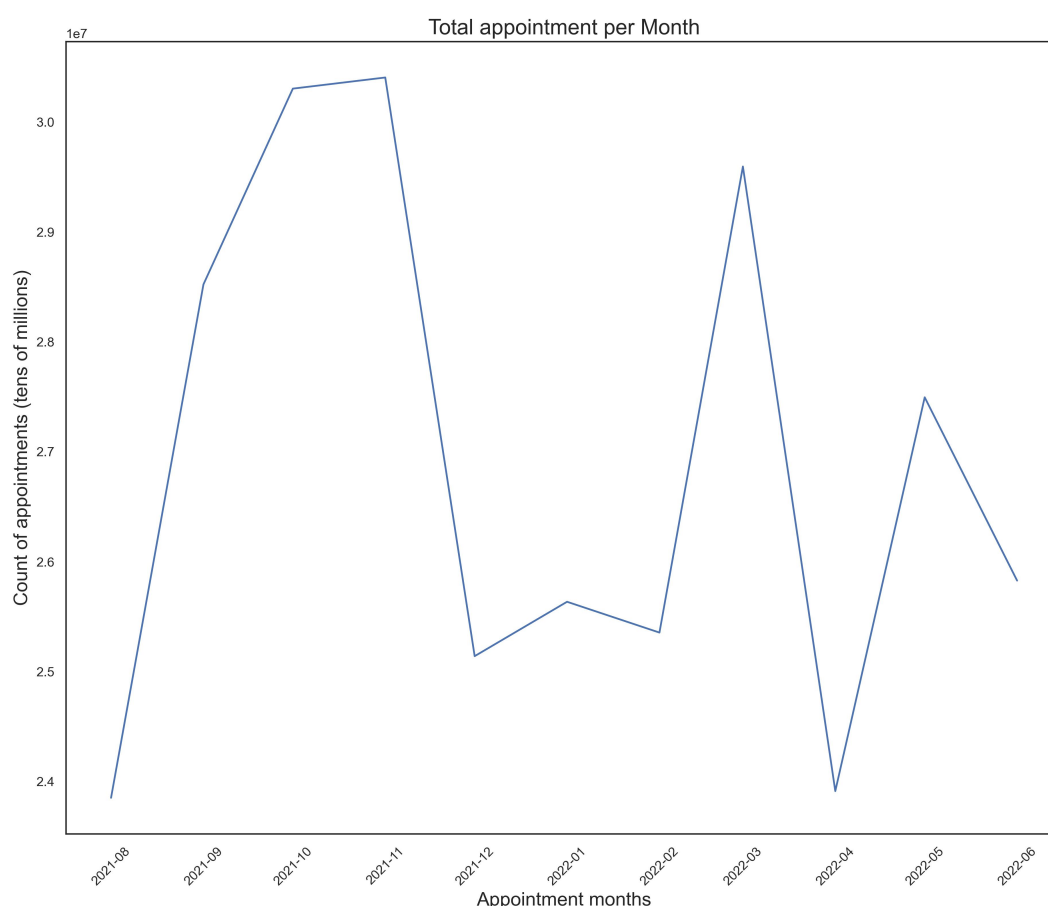
The Jupyter Notebook was structured as follows:

1. Import and validation: load datasets, check data types, and scan for null or ambiguous values
2. Exploration: descriptive statistics, unique value check
3. Trend analysis: group by time, season, and region
4. Capacity indicators: analyse staff type and size, service setting, utilisation rate of NHS facilities and fluctuation of several factors
5. Utilisation metrics: combine appointments fluctuation with utilisation of NHS capacity

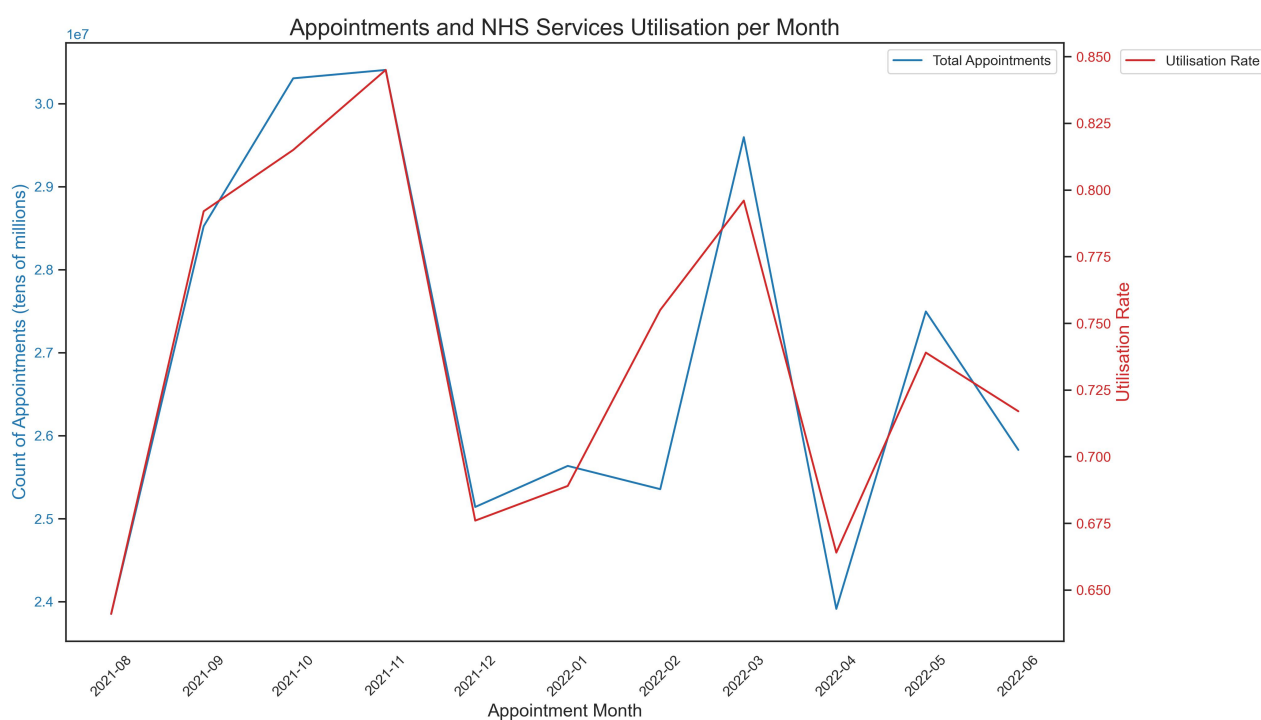
Twitter was valuable for Social media sentiment analysis which could reveal patient satisfaction, concerns about waiting times, or the perception of NHS services. Furthermore the features of retweeting and liking others message have an important multiplier effect. I find no relevant complaints associated with posts about the heal care system.

SUMMARY, RECOMMENDATIONS AND CHALLENGES

In summary, this analysis revealed key insights into the NHS resource allocation and usage:

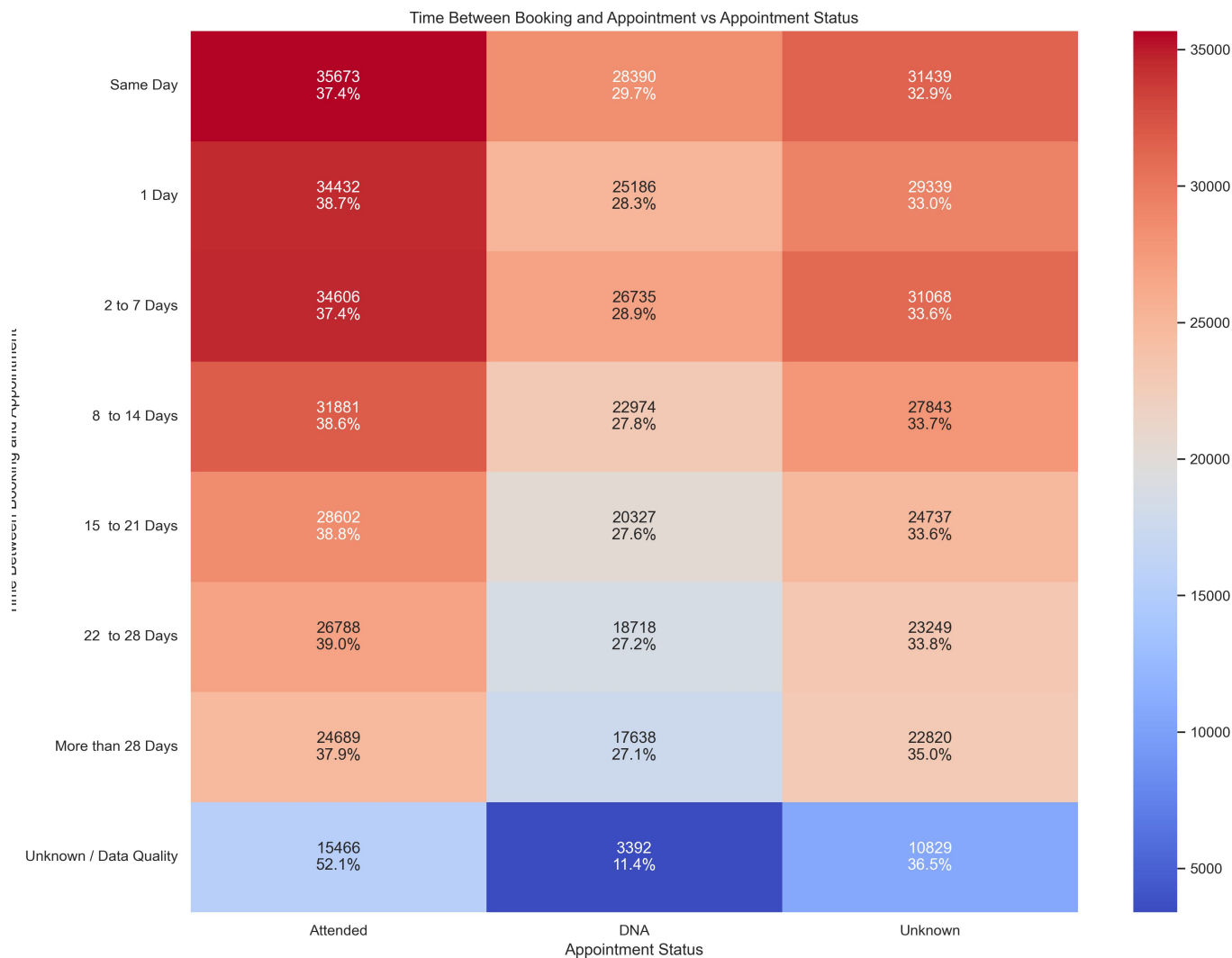


- Clear seasonal demand. Peaks in autumn and March suggest that resources must be weighted accordingly
- Face-to-face dominance. Indicates that in-person care remain central
- The utilisation rate compared to the total appointment show that during the peak moment there is a demand from the patients which is not satisfied by NHS



In the distribution of appointment duration, values in the '< 1 minutes or > 60 minutes' were detected as systemic inefficiency due to an incorrect communication for the right correct patient referral.

I combined the 'time between booking and appointment' and the 'appointment status' to look for a possible correlation but the 'unknown' data made it impossible to draw conclusion. This approach could be repeated in further analysis with better quality data.



To improve data collection and analysis I recommend a more rigorous data entry validation particularly for actual duration appointment field. Would be worth a further analysis repeating the current work for each region or ICB location. While time-consuming, could offer targeted improvement opportunities.

RECOMMENDATIONS

1. Dynamic staff models. Use peak-month forecasts to deploy staff where needed most
2. Leverage patients communication tools. Reduce DNAs through reminders or digital rebooking options
3. Expand use of remote modes. Promote telephone or video consultations when appropriate

The most important is to track data of the changes so we can monitor how the changes are perceived and used. Despite the challenges we are facing we support the NHS decision-makers for improvements and monitoring.